Post-merge validation and named-curve controls

2026-10-15 · Isogeny performance study · correctness receipts

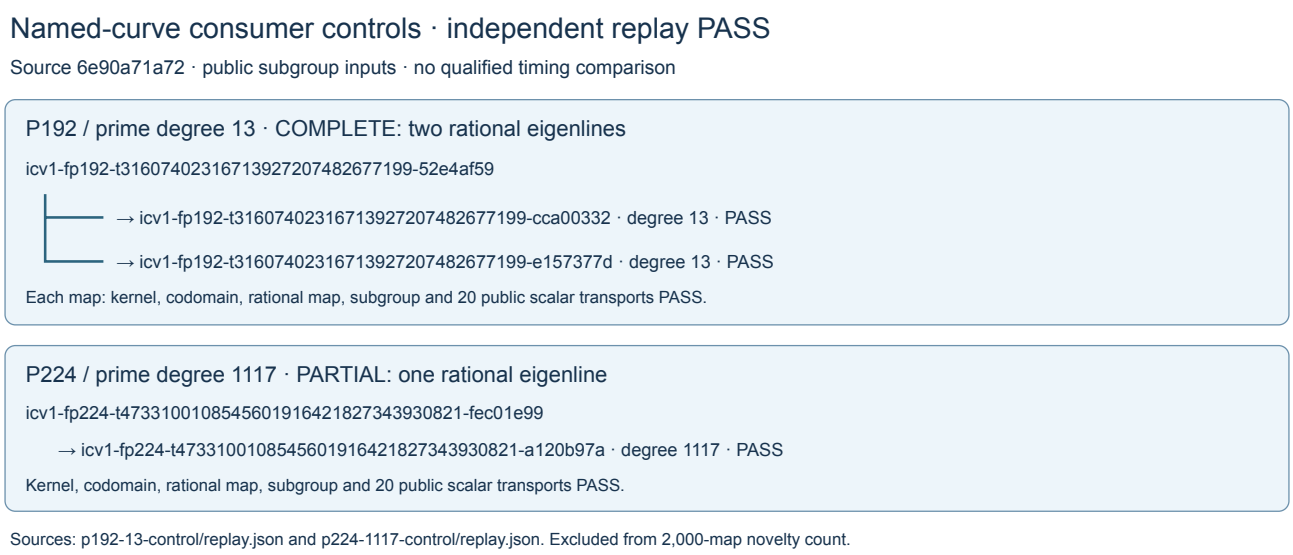
Source for post-merge tests: 0191f5b924765a0f6d7882f13787b7391413c685. The arithmetic implementation is unchanged from the final BMSS rung. Apple M4 Pro, 14 logical cores, 48 GiB, shared macOS host; actual compiler Homebrew rustc 1.93.1. No qualified wall timing is reported here.

Full root and standalone validation

Attempt	Passed	Failed	Result / boundary
Two test threads	4,094	1	117 existing ignored; Nagao deadline expired
Exact Nagao rerun	1	0	One Rayon worker; all assertions retained
Concurrent Rayon-one retry	Incomplete	Unknown	Interrupted; no full-suite verdict
Full serial Rayon-one suite	4,095	0	117 existing ignored; terminal PASS

The first failure explicitly reports `timed_out: true`. The exact test and solver source are unchanged. The successful serial run retains every test, assertion and deadline. The independent shell-exit receipt for that completed serial tool session was no longer retrievable on resumption; its complete terminal test verdict is retained. Full standalone release library, binary, integration and example suites also passed. Raw failed and interrupted receipts remain beside the successful run. Thread interference is a supported hypothesis, without a native isolation receipt proving its precise cause.

Dedicated-binary consumer controls



Both controls use the embedded public catalogue and independent verifier. Consumer build source: 6e90a71a72b7b7c3522bb0307ad1d25299bc98ce. P192/13 enumerates both eigenlines; P224/1117 reports one eigenline with PARTIAL coverage. Every map passes kernel, codomain, exact rational-map, subgroup and twenty public scalar-transport checks. These controls are excluded from the separate 2,000-additional-map requirement. They are correctness runs, not paired performance samples; a timing graph would have no admitted comparison.

P192 certificate SHA256: 7a1809d56cb774f681e06ecf1dad9e04d97abe70a3eb2738bff9d54affc861d1 P224 certificate SHA256: 6f1812ef5c90fabc34093c8b9545053997b55f851c44a9883bfa1ea2842d05fd

Complete commands, raw logs, maps, stderr, resource receipts and independent replay records are retained in this directory. Read README.md and FILES.json. The earlier frozen archives and their inventories are unchanged. Multi-CPU constructor comparisons and the remaining algorithm families remain open.